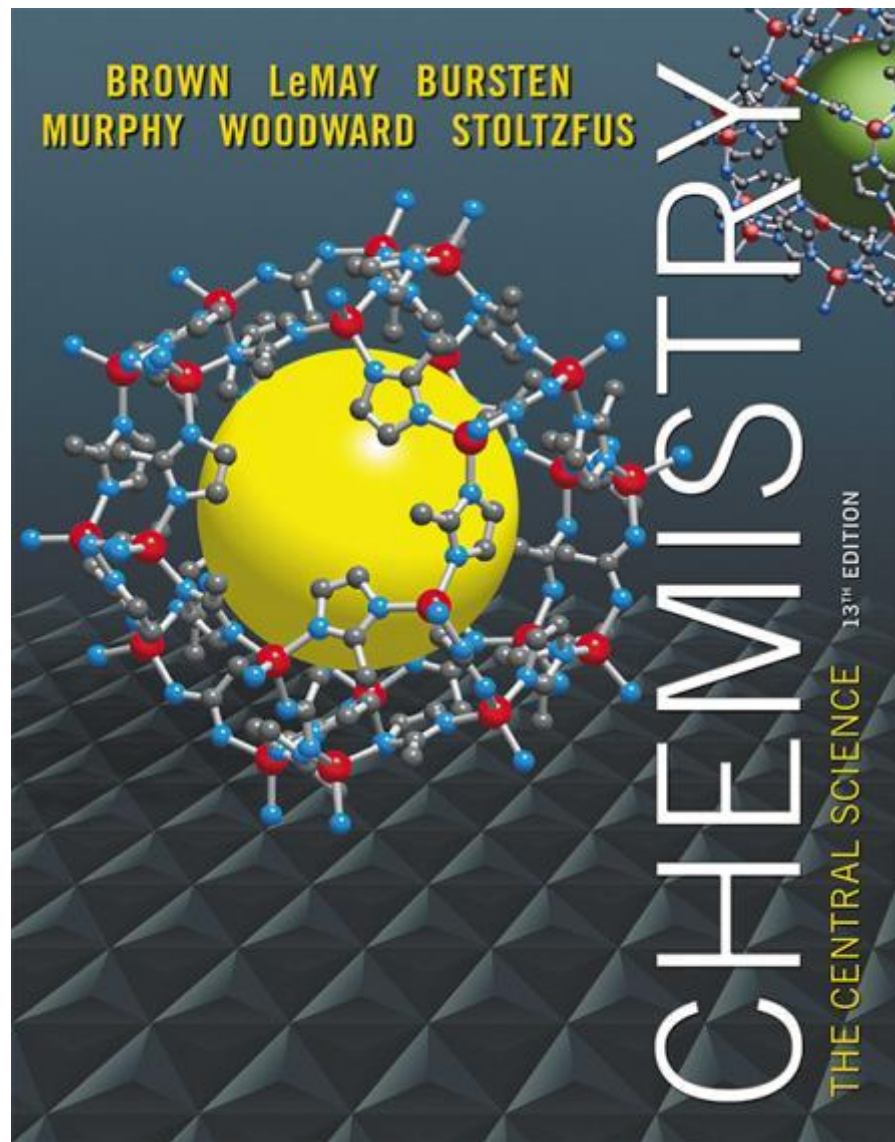


# Chapter 5. Organic and Biological Chemistry



**Reference:**

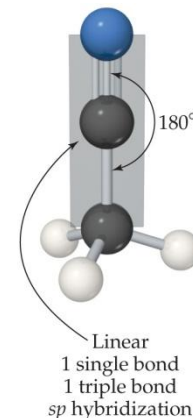
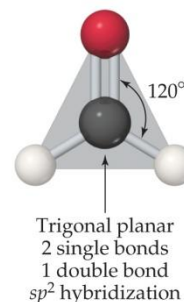
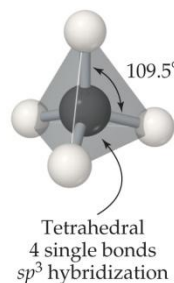
**Chapter 24**

**Organic and Biological Chemistry**

# Organic Chemistry and Biochemistry

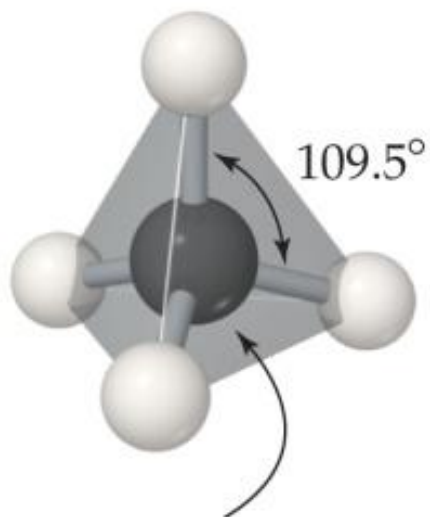
- Chapter focus: the molecules that bridge chemistry & biology
- Most common elements: C, H, O, N
- Organic chemistry: study of compounds containing carbon
- Biochemistry: the study of chemistry of living systems

# General Characteristics of Organic Molecules

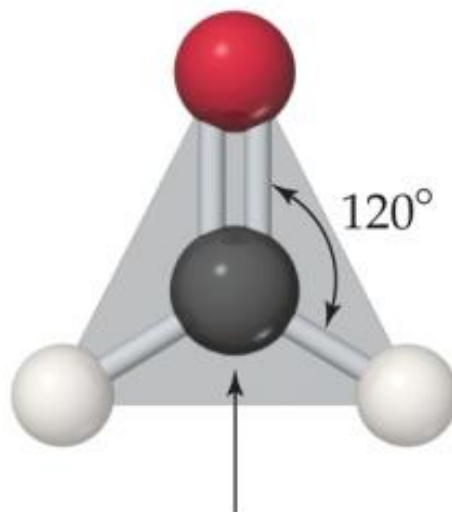


- Carbon makes four bonds.
- All single bonds: tetrahedral;  $sp^3$  hybridized
- One double bond: trigonal planar;  $sp^2$  hybridized
- One triple bond: linear;  $sp$  hybridized
- C—H are most common.
- C forms stable (strong) bonds with many elements, including C, H, O, N, and the halogens (F, Cl, Br, I).
- Groups of atoms that determine how an organic molecule reacts are called **functional groups**.

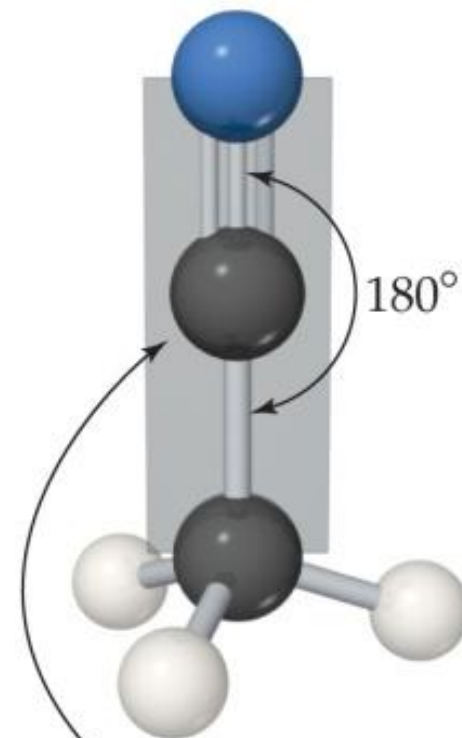
# General Characteristics of Organic Molecules



Tetrahedral  
4 single bonds  
 $sp^3$  hybridization



Trigonal planar  
2 single bonds  
1 double bond  
 $sp^2$  hybridization

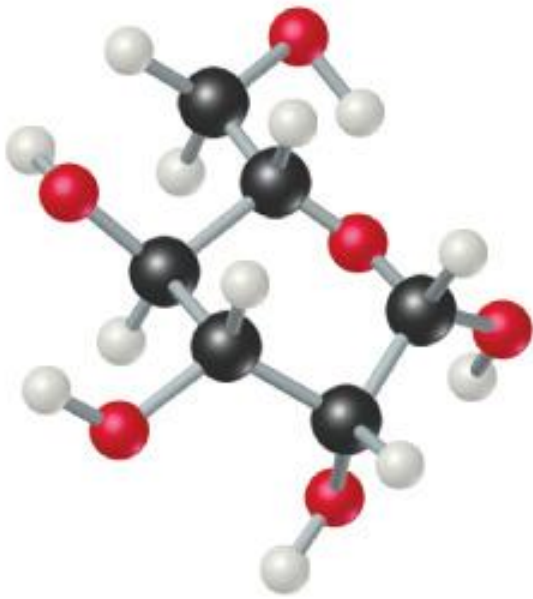


Linear  
1 single bond  
1 triple bond  
 $sp$  hybridization

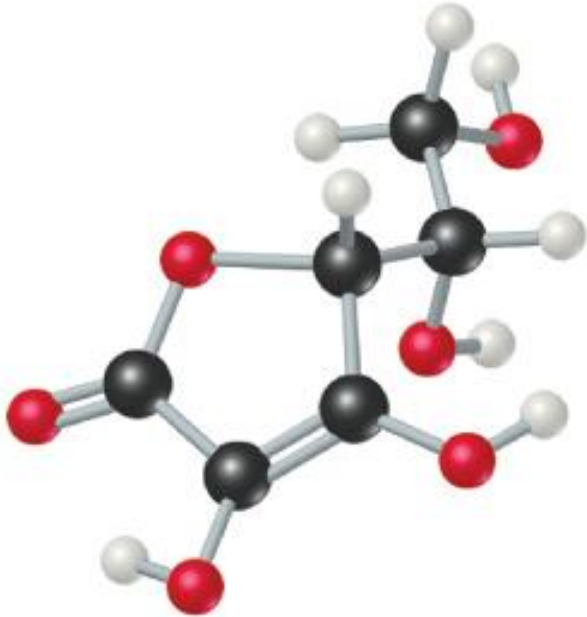
# Solubility

- Most prevalent bonds are C—C and C—H, which are nonpolar; solubility in water is low for many organic compounds.
- Organic molecules, such as **glucose** and **ascorbic acid**, that have polar groups are soluble in polar solvents.
- Molecules with long nonpolar regions and polar regions act as **surfactants**, bringing **nonpolar** material into aqueous solution (used in detergents and soaps).

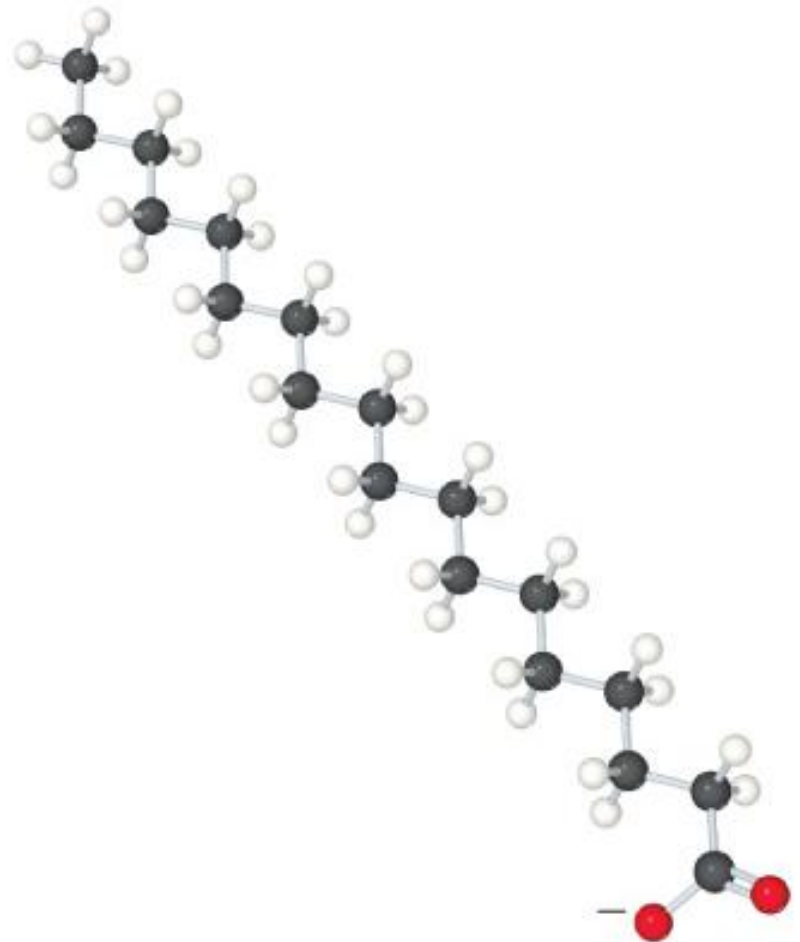
# Solubility



Glucose ( $\text{C}_6\text{H}_{12}\text{O}_6$ )



Ascorbic acid ( $\text{HC}_6\text{H}_7\text{O}_6$ )



Stearate ( $\text{C}_{17}\text{H}_{35}\text{COO}^-$ )

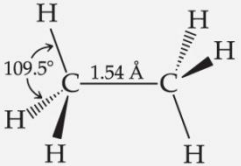
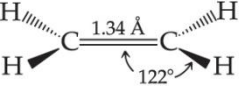
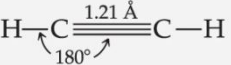
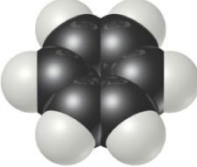
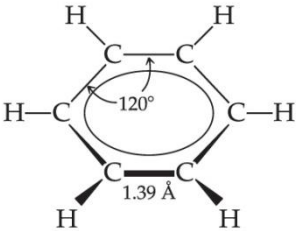
# Acid–Base Properties

- Many organic molecules contain acidic or basic functional groups.
- Carboxylic acids ( $\text{—COOH}$ ) are the most **common acids**.
- Amines ( $\text{—NH}_2$ ,  $\text{—NHR}$ , or  $\text{—NR}_2$ , where R is an organic group made up of C and H atoms) are the most **common bases**.

# Hydrocarbons

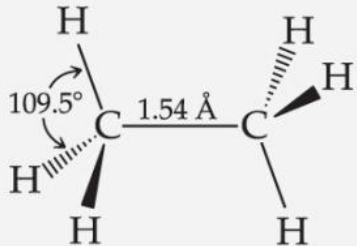
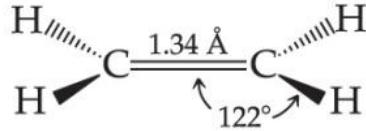
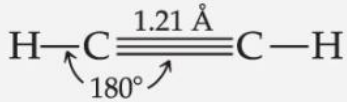
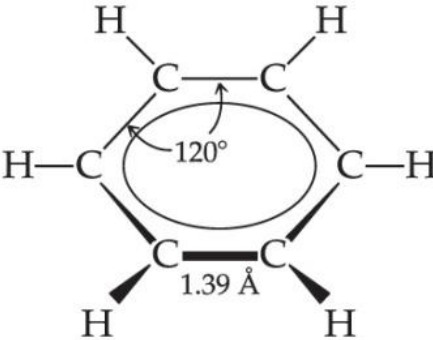
- Hydrocarbons consist of **ONLY** carbon and hydrogen.
- They are grouped based on the number of bonds between carbon atoms.
- There are four basic types of hydrocarbons:
  - Alkanes
  - Alkenes
  - Alkynes
  - Aromatic hydrocarbons

Table 24.1 The Four Hydrocarbon Types with Molecular Examples

| Type     | Example   |                            |                                                                                                                                                                           |
|----------|-----------|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Alkane   | Ethane    | $\text{CH}_3\text{CH}_3$   |     |
| Alkene   | Ethylene  | $\text{CH}_2=\text{CH}_2$  |     |
| Alkyne   | Acetylene | $\text{CH}\equiv\text{CH}$ |     |
| Aromatic | Benzene   | $\text{C}_6\text{H}_6$     |   |

# Hydrocarbons

Table 24.1 The Four Hydrocarbon Types with Molecular Examples

| Type     | Example   |                            |                                                                                                                                                                            |
|----------|-----------|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Alkane   | Ethane    | $\text{CH}_3\text{CH}_3$   |       |
| Alkene   | Ethylene  | $\text{CH}_2=\text{CH}_2$  |       |
| Alkyne   | Acetylene | $\text{CH}\equiv\text{CH}$ |       |
| Aromatic | Benzene   | $\text{C}_6\text{H}_6$     |   |

# Properties Common to Hydrocarbons

- Since they are nonpolar, they are insoluble in water but soluble in nonpolar solvents.
- Melting points and boiling points are determined by dispersion forces (low molar mass hydrocarbons are gases; moderate molar mass hydrocarbons are liquids; high molar mass hydrocarbons are solids).

Table 24.2 First Ten Members of the Straight-Chain Alkane Series

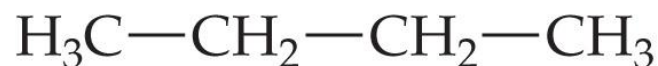
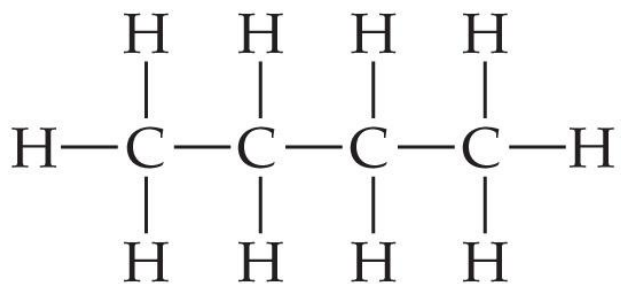
| Molecular Formula               | Condensed Structural Formula                                                                                                                                    | Name    | Boiling Point (°C) |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|--------------------|
| CH <sub>4</sub>                 | CH <sub>4</sub>                                                                                                                                                 | Methane | -161               |
| C <sub>2</sub> H <sub>6</sub>   | CH <sub>3</sub> CH <sub>3</sub>                                                                                                                                 | Ethane  | -89                |
| C <sub>3</sub> H <sub>8</sub>   | CH <sub>3</sub> CH <sub>2</sub> CH <sub>3</sub>                                                                                                                 | Propane | -44                |
| C <sub>4</sub> H <sub>10</sub>  | CH <sub>3</sub> CH <sub>2</sub> CH <sub>2</sub> CH <sub>3</sub>                                                                                                 | Butane  | -0.5               |
| C <sub>5</sub> H <sub>12</sub>  | CH <sub>3</sub> CH <sub>2</sub> CH <sub>2</sub> CH <sub>2</sub> CH <sub>3</sub>                                                                                 | Pentane | 36                 |
| C <sub>6</sub> H <sub>14</sub>  | CH <sub>3</sub> CH <sub>2</sub> CH <sub>2</sub> CH <sub>2</sub> CH <sub>2</sub> CH <sub>3</sub>                                                                 | Hexane  | 68                 |
| C <sub>7</sub> H <sub>16</sub>  | CH <sub>3</sub> CH <sub>2</sub> CH <sub>2</sub> CH <sub>2</sub> CH <sub>2</sub> CH <sub>2</sub> CH <sub>3</sub>                                                 | Heptane | 98                 |
| C <sub>8</sub> H <sub>18</sub>  | CH <sub>3</sub> CH <sub>2</sub> CH <sub>2</sub> CH <sub>2</sub> CH <sub>2</sub> CH <sub>2</sub> CH <sub>2</sub> CH <sub>3</sub>                                 | Octane  | 125                |
| C <sub>9</sub> H <sub>20</sub>  | CH <sub>3</sub> CH <sub>2</sub> CH <sub>2</sub> CH <sub>2</sub> CH <sub>2</sub> CH <sub>2</sub> CH <sub>2</sub> CH <sub>2</sub> CH <sub>3</sub>                 | Nonane  | 151                |
| C <sub>10</sub> H <sub>22</sub> | CH <sub>3</sub> CH <sub>2</sub> CH <sub>2</sub> CH <sub>2</sub> CH <sub>2</sub> CH <sub>2</sub> CH <sub>2</sub> CH <sub>2</sub> CH <sub>2</sub> CH <sub>3</sub> | Decane  | 174                |

# Uses of Some Simple Alkanes

- Methane ( $\text{CH}_4$ ): in natural gas (heating fuel)
- Propane ( $\text{C}_3\text{H}_8$ ): in bottled gas (heating and cooking fuel)
- Butane ( $\text{C}_4\text{H}_{10}$ ): in disposable lighters and fuel canisters for camping
- Alkanes with 5 to 12 C atoms: gasoline

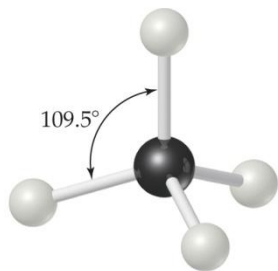
# Methods for Writing Formulas

- There are a few common methods for writing the structures in organic chemistry.
- Structural formulas show how atoms are bonded to each other.
- Condensed structural formulas don't show all C—H; they condense them to groupings, like CH<sub>3</sub>.



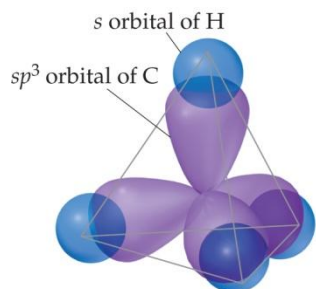
or



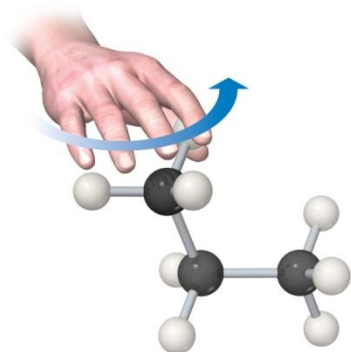


# Structure of Alkanes

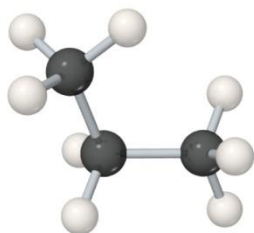
- Carbons in alkanes are  $sp^3$  hybridized, tetrahedral, and have  $109.5^\circ$  bond angles.



- In the straight chain form, all carbons connect in a continuous chain.



- C can make four bonds, so it is possible for a carbon atom to bond to three or four C atoms, making a branched alkane.
- Compounds with the same molecular formula but different connections of atoms are called **structural isomers**, as seen on the next slide.



# Structural Isomers

Table 24.3 Isomers of  $C_4H_{10}$  and  $C_5H_{12}$

| Systematic Name<br>(Common Name)    | Structural Formula                                                                                                                                                                                                     | Condensed Structural<br>Formula                                                | Space-filling<br>Model                                                                | Melting<br>Point ( $^{\circ}C$ ) | Boiling<br>Point ( $^{\circ}C$ ) |
|-------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|----------------------------------|----------------------------------|
| Butane<br>( <i>n</i> -butane)       | $  \begin{array}{cccc}  H & H & H & H \\    &   &   &   \\  H-C & -C & -C & -C-H \\    &   &   &   \\  H & H & H & H  \end{array}  $                                                                                   | $CH_3CH_2CH_2CH_3$                                                             |    | -138                             | -0.5                             |
| 2-Methylpropane<br>(isobutane)      | $  \begin{array}{ccccc}  & H & & H & \\  &   & &   & \\  H & -C & - & C & -H \\  &   & &   & \\  & H & & H & \\  & & &   & \\  & & & H-C-H \\  & & &   \\  & & & H  \end{array}  $                                     | $  \begin{array}{c}  CH_3-CH-CH_3 \\    \\  CH_3  \end{array}  $               |    | -159                             | -12                              |
| Pentane<br>( <i>n</i> -pentane)     | $  \begin{array}{cccccc}  H & H & H & H & H \\    &   &   &   &   \\  H-C & -C & -C & -C & -C-H \\    &   &   &   &   \\  H & H & H & H & H  \end{array}  $                                                            | $CH_3CH_2CH_2CH_2CH_3$                                                         |    | -130                             | +36                              |
| 2-Methylbutane<br>(isopentane)      | $  \begin{array}{ccccc}  & H & & & \\  &   & & H & \\  H & -C & -H & & \\  &   & &   & \\  H & -C & -C & -C & -H \\  &   & &   & \\  & H & & H & \\  & & &   & \\  & & & H-C-H \\  & & &   \\  & & & H  \end{array}  $ | $  \begin{array}{c}  CH_3 \\    \\  CH_3-CH-CH_2-CH_3  \end{array}  $          |   | -160                             | +28                              |
| 2,2-Dimethylpropane<br>(neopentane) | $  \begin{array}{ccccc}  & & H & & \\  & &   & & \\  H & & C & & H \\  & &   & & \\  H & -C & - & C & -H \\  & &   & & \\  & & H & & \\  & &   & & \\  & & H-C-H \\  & &   \\  & & H  \end{array}  $                   | $  \begin{array}{c}  CH_3 \\    \\  CH_3-C-CH_3 \\    \\  CH_3  \end{array}  $ |  | -16                              | +9                               |

Name each substituent: Side chains that are based on alkanes are called alkyl groups.

#### First Ten Members of the Straight-Chain Alkane Series

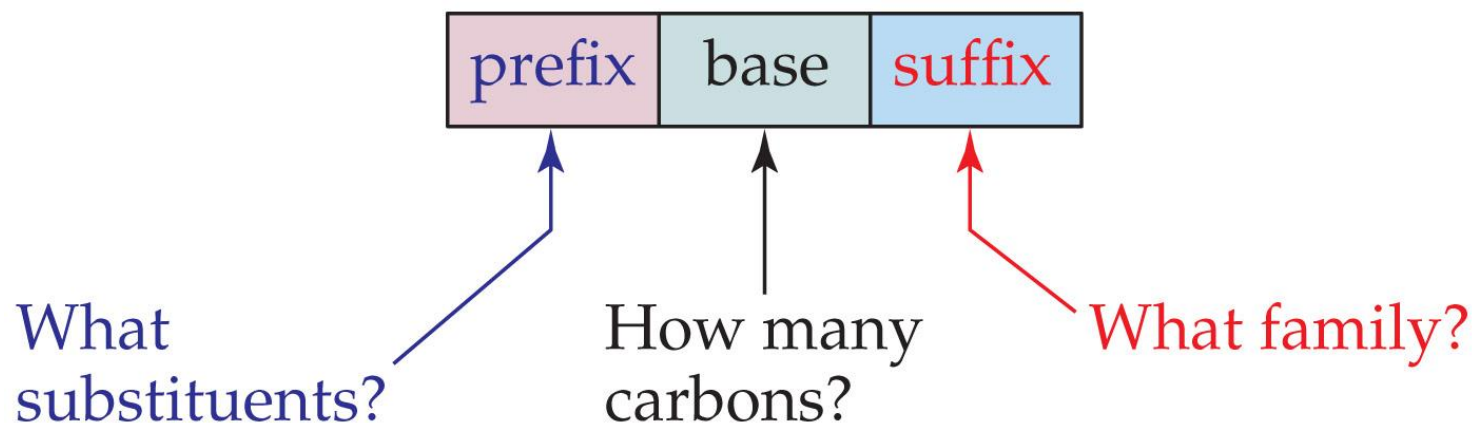
| Condensed Structural Formula                                                                                     | Name    |
|------------------------------------------------------------------------------------------------------------------|---------|
| $\text{CH}_4$                                                                                                    | Methane |
| $\text{CH}_3\text{CH}_3$                                                                                         | Ethane  |
| $\text{CH}_3\text{CH}_2\text{CH}_3$                                                                              | Propane |
| $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_3$                                                                   | Butane  |
| $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3$                                                        | Pentane |
| $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3$                                             | Hexane  |
| $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3$                                  | Heptane |
| $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3$                       | Octane  |
| $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3$            | Nonane  |
| $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3$ | Decane  |

Table 24.4 Condensed Structural Formulas and Common Names for Several Alkyl Groups

| Group                                                                                                  | Name               |
|--------------------------------------------------------------------------------------------------------|--------------------|
| $\text{CH}_3\text{—}$                                                                                  | Methyl             |
| $\text{CH}_3\text{CH}_2\text{—}$                                                                       | Ethyl              |
| $\text{CH}_3\text{CH}_2\text{CH}_2\text{—}$                                                            | Propyl             |
| $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{—}$                                                 | Butyl              |
| $  \begin{array}{c}  \text{CH}_3 \\    \\  \text{HC—} \\    \\  \text{CH}_3  \end{array}  $            | Isopropyl          |
| $  \begin{array}{c}  \text{CH}_3 \\    \\  \text{CH}_3\text{—C—} \\    \\  \text{CH}_3  \end{array}  $ | <i>tert</i> -Butyl |

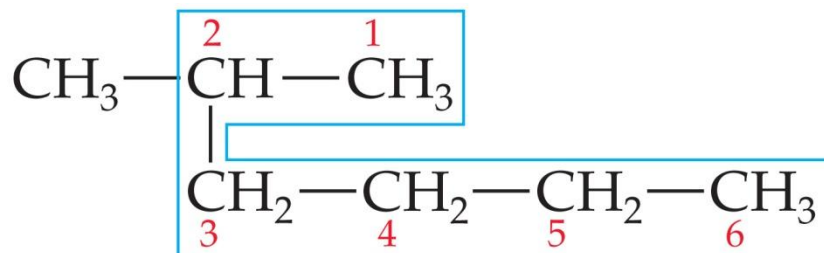
# Systematic Nomenclature of Organic Compounds

- There are three parts to a compound name:
  - Base: This tells how many carbons are in the longest continuous chain.
  - Suffix: This tells what type of compound it is.
  - Prefix: This tells what groups are attached to the chain.



# How to Name a Compound

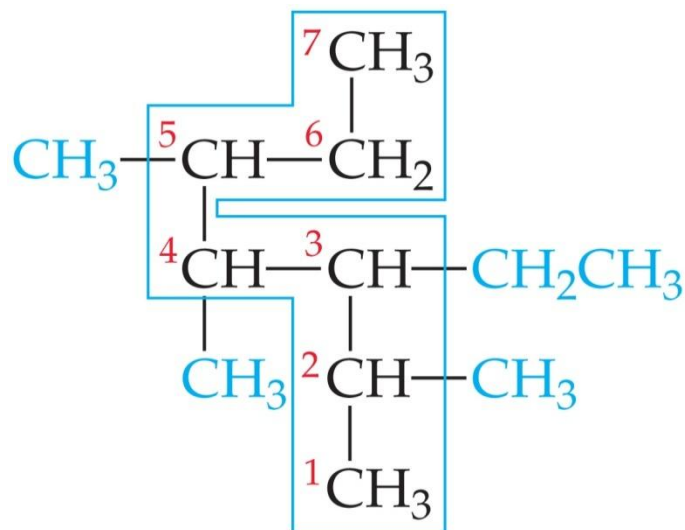
1. Find the longest continuous chain of C atoms, and use this as the base name.
2. Number the chain from the end nearest the first substituent encountered.
3. Name each substituent. (Side chains that are based on alkanes are called alkyl groups.)



2-Methylhexane

# How to Name a Compound

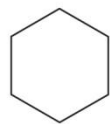
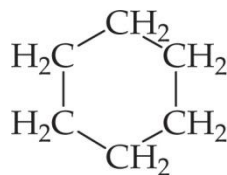
4. Begin the name with the number(s) on the C atom(s) to which each substituent is bonded.
5. When two or more substituents are present, list them alphabetically.



3-Ethyl-2,4,5-trimethylheptane

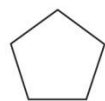
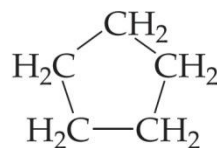
# Cycloalkanes

- Alkanes that form rings or cycles
- Possible with at least three C atoms, but  $sp^3$  hybridization requires  $109.5^\circ$  angles—not a very stable molecule.
- Four-C ring is also not very stable.
- Five-C and more have room for proper bond angle.
- Naming: add *cyclo-* as a prefix to alkane name.



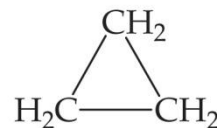
Cyclohexane

Each vertex  
represents one  
CH<sub>2</sub> group



Cyclopentane

Five vertices =  
five CH<sub>2</sub> groups

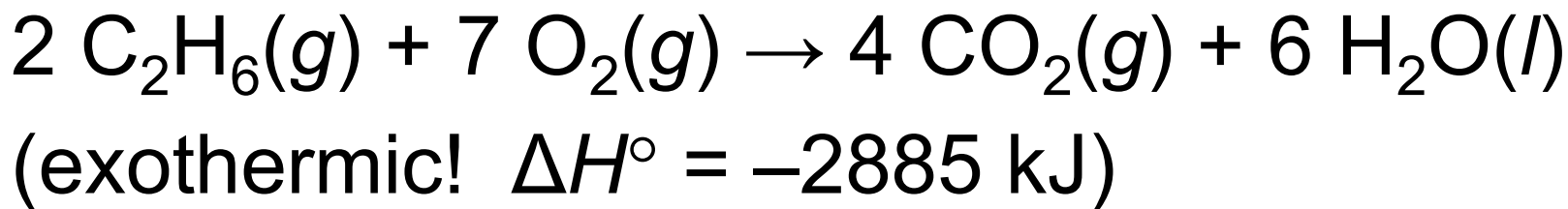


Cyclopropane

Three vertices =  
three CH<sub>2</sub> groups

# Reactions of Alkanes

- Alkanes are relatively unreactive due to the lack of polarity and presence of only C—C and C—H  $\sigma$  bonds, which are very stable.
- They do not react with acids, bases, or oxidizing agents.
- However, the most important reaction observed is combustion:



# Saturated vs. Unsaturated

- Hydrocarbons with single bonds only are called saturated hydrocarbons. These are the alkanes.
- Alkenes, alkynes, and aromatic hydrocarbons have fewer hydrogen atoms than alkanes with the same number of carbon atoms. They are called unsaturated hydrocarbons.
- Unsaturated hydrocarbons are more reactive than saturated hydrocarbons.

# Alkenes

- Contain at least one C=C bond
- No free rotation about the double bond
- Naming: longest chain must include BOTH carbon atoms that share the double bond; end name in *-ene*; lowest number possible given to double-bond carbon atoms; isomers also indicated.

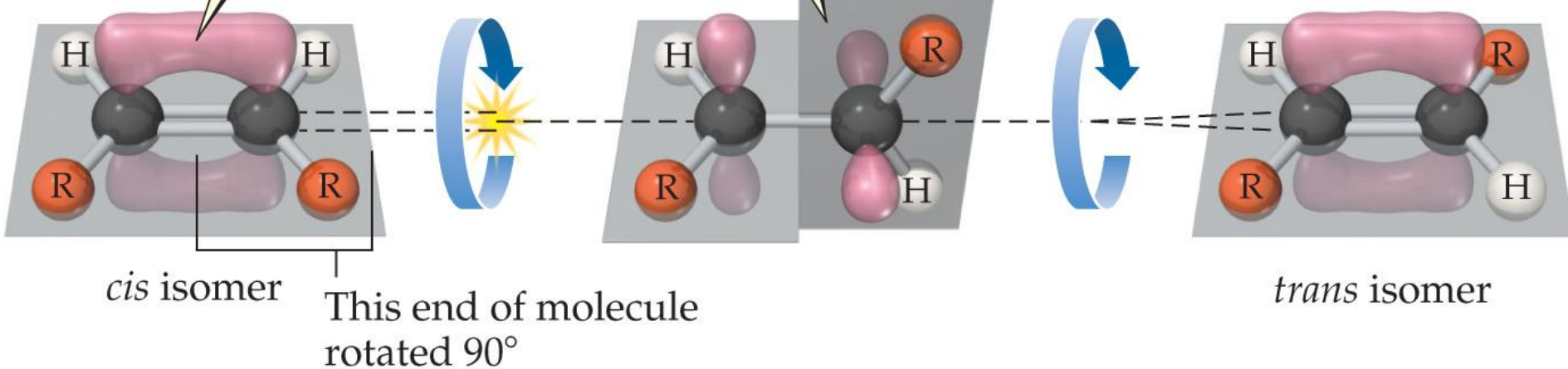
<https://www.chemistryworld.com/news/synthetic-strategies-overcome-bredts-rule-unlocking-complex-bridgehead-alkenes/4020482.article>

# Alkenes

Overlapping  $p$  orbitals create one  $\pi$  bond

Rotation around double bond requires considerable energy to break  $\pi$  bond

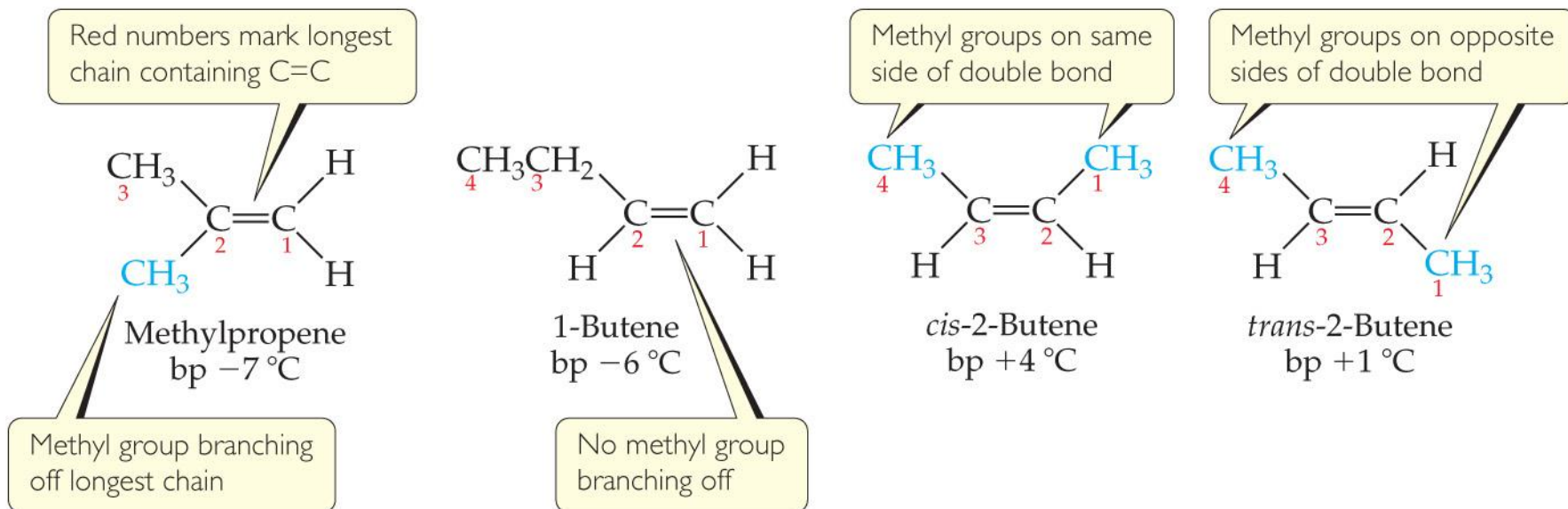
Rotation has destroyed orbital overlap



# Geometric Isomers

- Since there is no free rotation around the double bond, the direction of the longest chain can differ for four or more C atoms.
- Compounds that have all atoms connected to the same atoms but differ in three-dimensional arrangement are **geometric isomers**.
- Alkenes have cis (same side of the double bond) or trans (opposite side of the double bond) isomers.

# Geometric Isomers

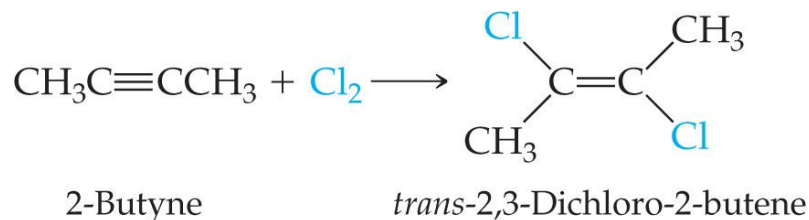
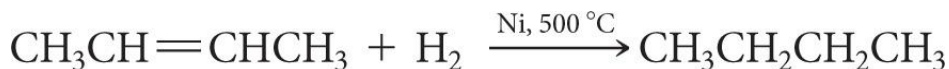
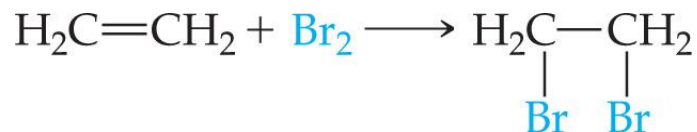


# Alkynes

- Contain at least one  $\text{C}\equiv\text{C}$
- Unsaturated
- Naming: longest continuous chain containing both carbon atoms in the triple bond; name ends in *-yne* (instead of *-ane* or *-ene*); give C atoms in triple bond lowest number.

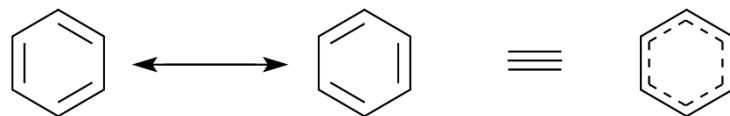
# Addition Reactions of Alkenes and Alkynes

- Add atoms to the double or triple bond, making it saturated or more saturated
- $\pi$  bonds are broken and electrons form  $\sigma$  bonds to added atoms.
- Work with  $H_2$  (hydrogenation),  $HX$  (hydrogen halides or water), or  $X_2$  (halogenation)

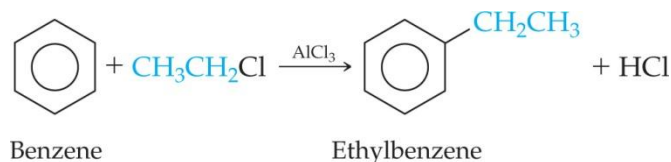
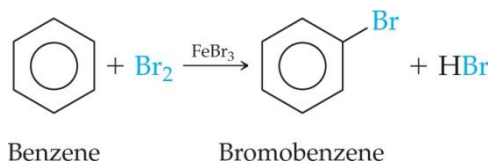
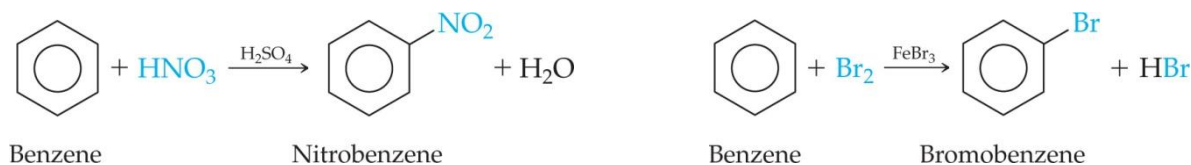


# Aromatic Hydrocarbons

- Aromatic hydrocarbons have six-membered rings containing localized and delocalized electrons.
- The  $\pi$  ring is much more stable than a  $\pi$  bond. So, aromatic hydrocarbons are much less reactive than alkenes and alkynes.

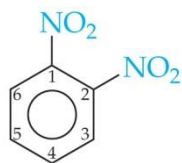


- They undergo substitution reactions rather than addition reactions: groups replace H on a ring (e.g., nitration, halogenation, alkylation).

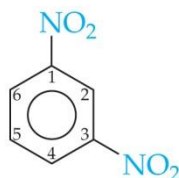


# Aromatic Nomenclature

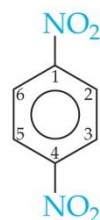
- Many aromatic hydrocarbons are known by their common names.
- Others are named as derivatives of benzene.
- Substitution positions for two substituents:  
1,2 = *ortho*-; 1,3 = *meta*-; 1,4 = *para*-



*ortho*-Dinitrobenzene  
1,2-Dinitrobenzene  
mp 118 °C



*meta*-Dinitrobenzene  
1,3-Dinitrobenzene  
mp 90 °C



*para*-Dinitrobenzene  
1,4-Dinitrobenzene  
mp 174 °C



Benzene



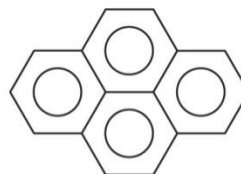
Naphthalene



Anthracene



Toluene  
(Methylbenzene)



Pyrene

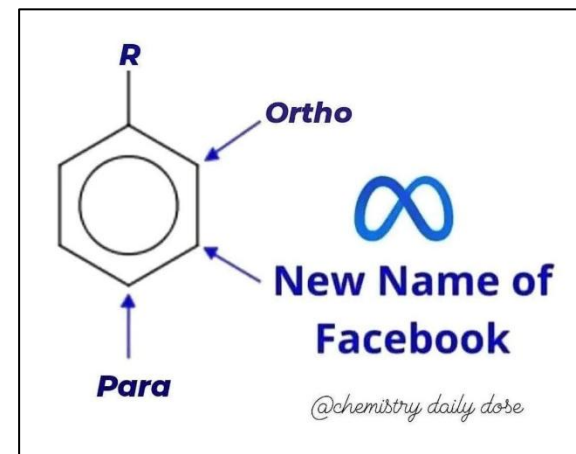


Table 24.6 Common Functional Groups

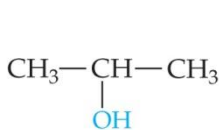
| Functional Group           | Compound Type                 | Suffix or Prefix | Structural Formula           | Example<br>Ball-and-stick Model | Systematic Name<br>(common name)     |
|----------------------------|-------------------------------|------------------|------------------------------|---------------------------------|--------------------------------------|
|                            | Alkene                        | -ene             |                              |                                 | Ethene<br>(Ethylene)                 |
| $\text{—C}\equiv\text{C—}$ | Alkyne                        | -yne             | $\text{H—C}\equiv\text{C—H}$ |                                 | Ethyne<br>(Acetylene)                |
|                            | Alcohol                       | -ol              |                              |                                 | Methanol<br>(Methyl alcohol)         |
|                            | Ether                         | ether            |                              |                                 | Dimethyl ether                       |
| <br>(X = halogen)          | Alkyl halide<br>or haloalkane | -ide             |                              |                                 | Chloromethane<br>(Methyl chloride)   |
|                            | Amine                         | -amine           |                              |                                 | Ethylamine                           |
|                            | Aldehyde                      | -al              |                              |                                 | Ethanal<br>(Acetaldehyde)            |
|                            | Ketone                        | -one             |                              |                                 | Propanone<br>(Acetone)               |
|                            | Carboxylic acid               | -oic acid        |                              |                                 | Ethanoic acid<br>(Acetic acid)       |
|                            | Ester                         | -oate            |                              |                                 | Methyl ethanoate<br>(Methyl acetate) |
|                            | Amide                         | -amide           |                              |                                 | Ethanamide<br>(Acetamide)            |

# Functional Groups

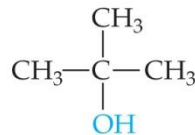
- The chemistry of an organic molecule is largely determined by the functional groups it contains.
- R represents the alkyl portion (C,H) of an organic molecule.

# Alcohols

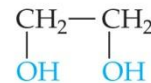
- Alcohols contain one or more —OH group (the alcohol group or the hydroxyl group).
- They are named from the parent hydrocarbon; the suffix is changed to **-ol** and a number designates the carbon to which the —OH group is attached.



2-Propanol  
Isopropyl alcohol;  
rubbing alcohol



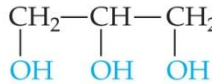
2-Methyl-2-propanol  
*t*-Butyl alcohol



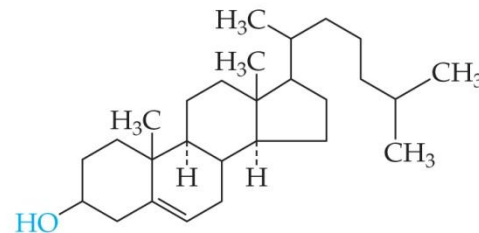
1,2-Ethanediol  
Ethylene glycol



Phenol



1,2,3-Propanetriol  
Glycerol; glycerin



2,15-dimethyl-14-(1,5-dimethylhexyl)tetracyclo[8.7.0.0<sup>2,7</sup>.0<sup>11,15</sup>]heptacos-7-en-5-ol  
Cholesterol

# Properties and Uses of Alcohols

- Polar molecules: lead to water solubility and higher boiling points
- Methanol: used as a fuel additive
- Ethanol: in alcoholic beverages
- Ethylene glycol: in antifreeze
- Glycerol: cosmetic skin softener and food moisturizer
- Phenol: making plastics and dyes; topical anesthetic in throat sprays
- Cholesterol: important biomolecule in membranes, but can precipitate and form gallstones or block blood vessels

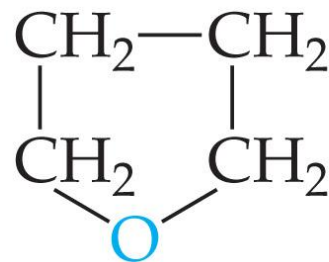


# Ethers

- $R-O-R$
- Formed by dehydration between alcohol molecules
- Not very reactive (except combustion)
- Used as solvents for organic reactions.



Diethyl ether



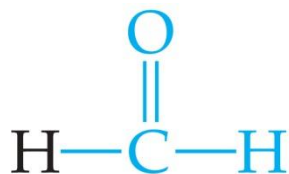
Tetrahydrofuran (THF)

# Functional Groups Containing the Carbonyl Group

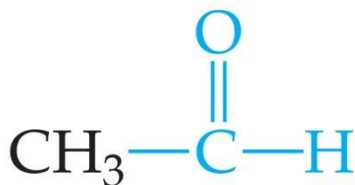
- The carbonyl group is  $\text{C}=\text{O}$ .
- Functional groups containing  $\text{C}=\text{O}$ :
  - Aldehydes
  - Ketones
  - Carboxylic acids
  - Esters
  - Amides

# Aldehydes and Ketones

- Aldehydes have at least one hydrogen atom attached to the carbonyl carbon atom.



Methanal  
Formaldehyde

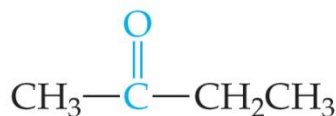


Ethanal  
Acetaldehyde

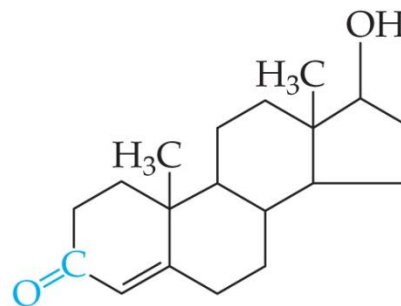
- Ketones have two R groups attached to the carbonyl carbon atom.



Propanone  
Acetone



2-Butanone  
Methyl ethyl ketone



Testosterone

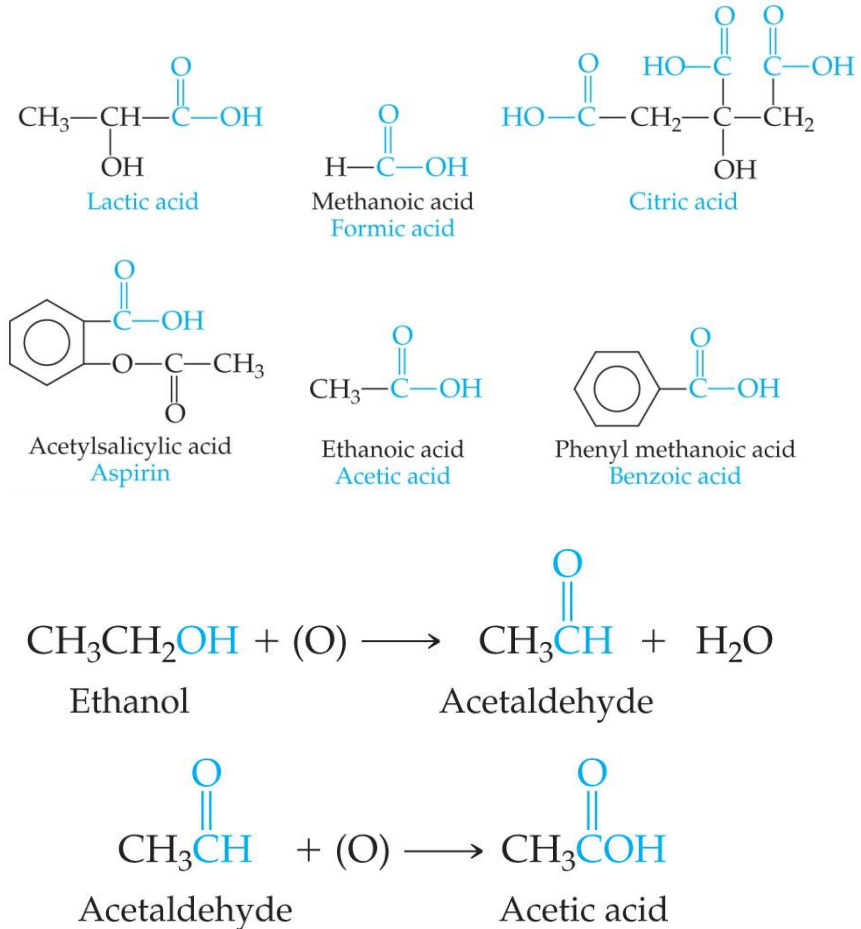
# Important Aldehydes and Ketones

- Many aldehydes are natural flavorings: vanilla, cinnamon, spearmint, and caraway are from aldehydes.
- Ketones are used extensively as solvents; the most important solvent other than water is acetone, which dissolves in water and dissolves many organic compounds.

<https://en.wikipedia.org/wiki/Vanillin>

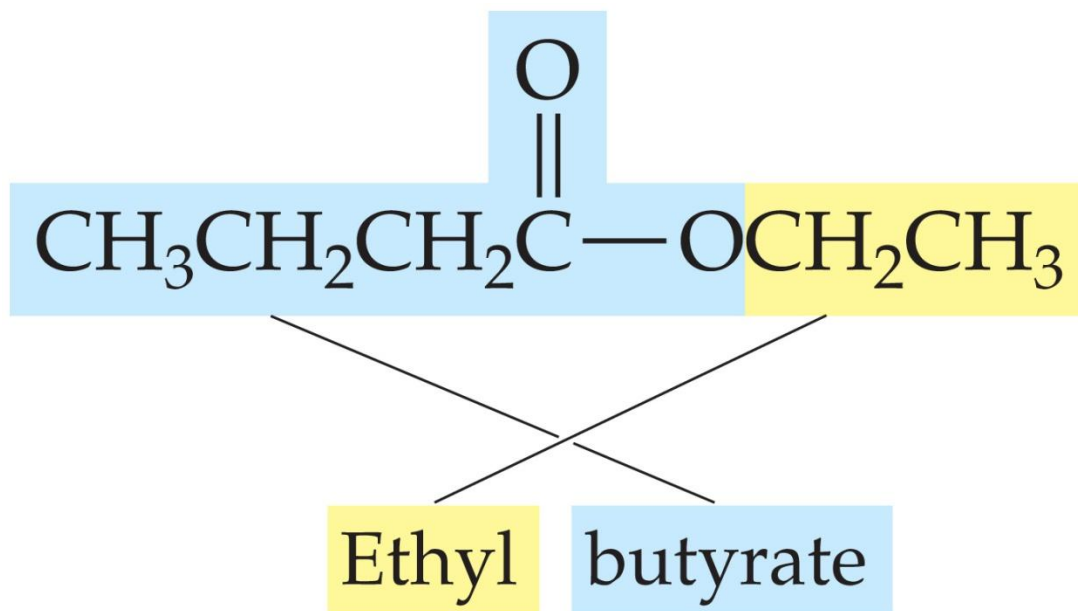
# Carboxylic Acids

- Structure: hydroxyl group bonded to the carbonyl group
- H on the hydroxyl group is weakly acidic.
- Important in manufacturing polymers for films, fibers, and paints
- Oxidation product of alcohols (some make aldehydes)



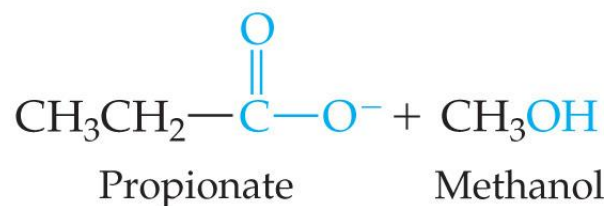
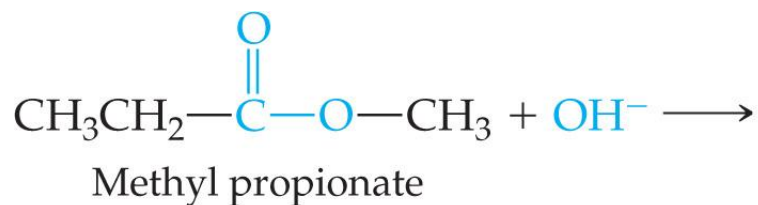
# Esters

- Esters are the products of reactions between carboxylic acids and alcohols.
- They are found in many fruits and perfumes.
- Naming: name the alcohol part as an alkyl name; separate word is the acid part as an *-ate* anion.



# Decomposition of Esters

- Heating an ester in the presence of an acid catalyst and water can decompose the ester. (This is the reverse reaction of the preparation of an ester; it is an equilibrium.)
- Heating an ester in the presence of a base results in **saponification** (making soap).



# Nitrogen Containing Organic Compounds

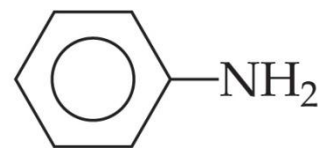
- Amines are organic derivatives of ammonia ( $\text{NH}_3$ ). One, two, or all three H atoms can be replaced by R groups (the same or different R groups).



Ethylamine



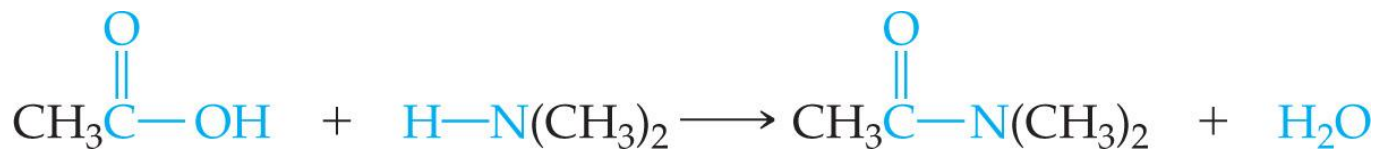
Trimethylamine



Phenylamine

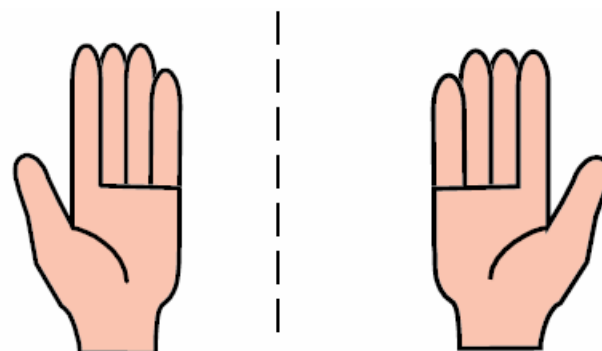
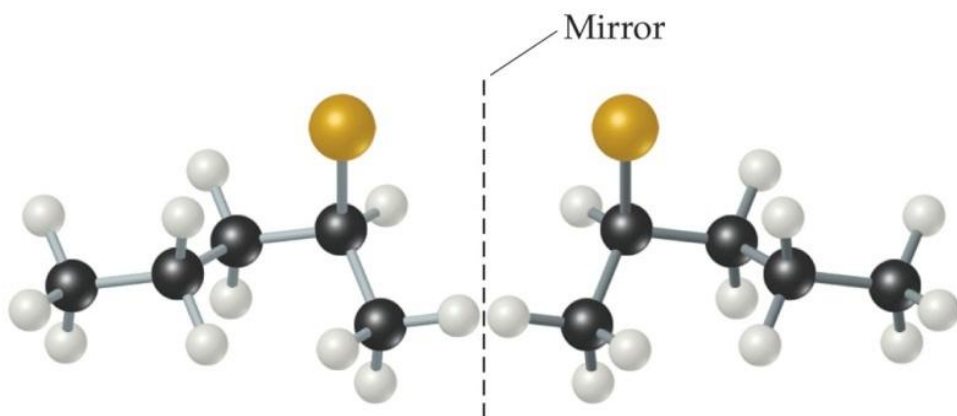
Aniline

- If H in  $\text{NH}_3$  or an amine is replaced by a carbonyl group (N directly attached to  $\text{C}=\text{O}$ ), an **amide** is formed.



# Chirality

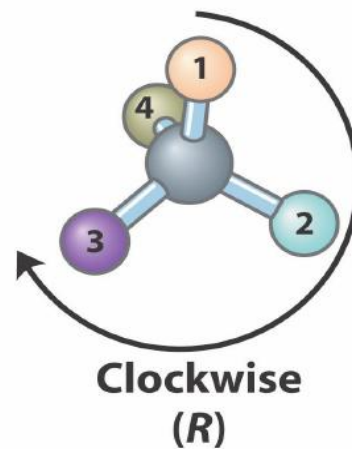
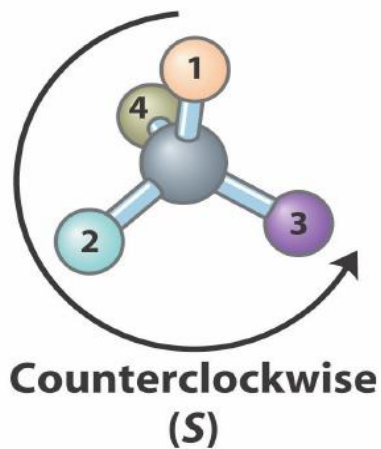
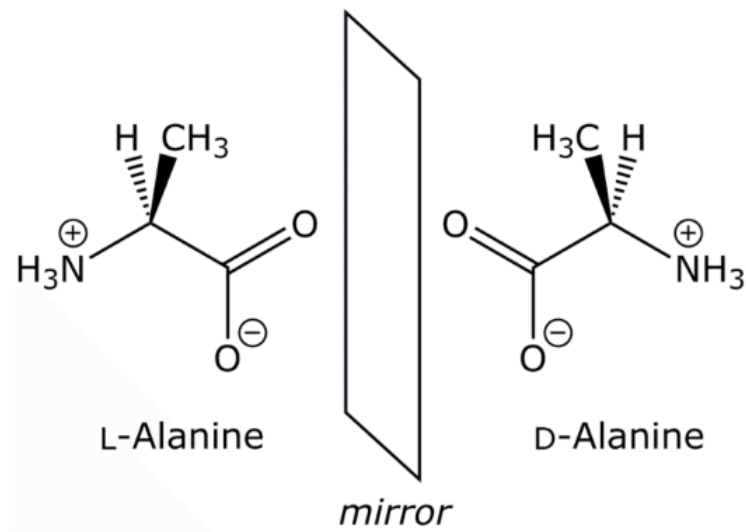
- Carbons with four different groups attached to them are **chiral**.
- These are optical isomers, or **enantiomers**.
- Enantiomers have the same physical and chemical properties when they react with **nonchiral** reagents.
- Enantiomers rotate plane-polarized light in opposite directions.



# Chiral Carbons Are Present in Most Biological Molecules

- Carbon atoms that are bound to **four** different atoms or groups are said to be asymmetric (**chiral**)
- The bonds formed by an asymmetric carbon can be arranged in two different mirror images (**stereoisomers**) of each other
- **Stereoisomers** are either right-handed or left-handed and typically have completely different biological activities
- Asymmetric carbons are key features of **amino acids**, **carbohydrates**, and **nucleic acids**
- Biomolecules are **stereospecific**, because they also are chiral and their binding sites may be recognized by different enantiomers.

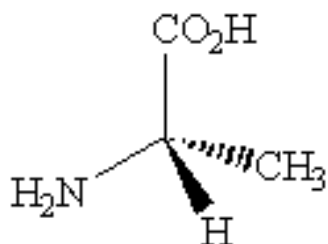
# Chirality



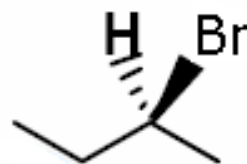
# Chirality

## R/S nomenclature for chiral center

- Locate the chiral center, identify its four substituents and assign priority from 1 (highest) to 4 (lowest) to each substituent.
- Orient the molecule so that the group of lowest priority (4) is directed away from you.
- Read the three groups projecting toward you in order from highest (1) to lowest priority (3).
- If the groups are read clockwise, the configuration is **R** (right); if they are read counterclockwise, the configuration is **S** (left).



**(S) - Alanine**

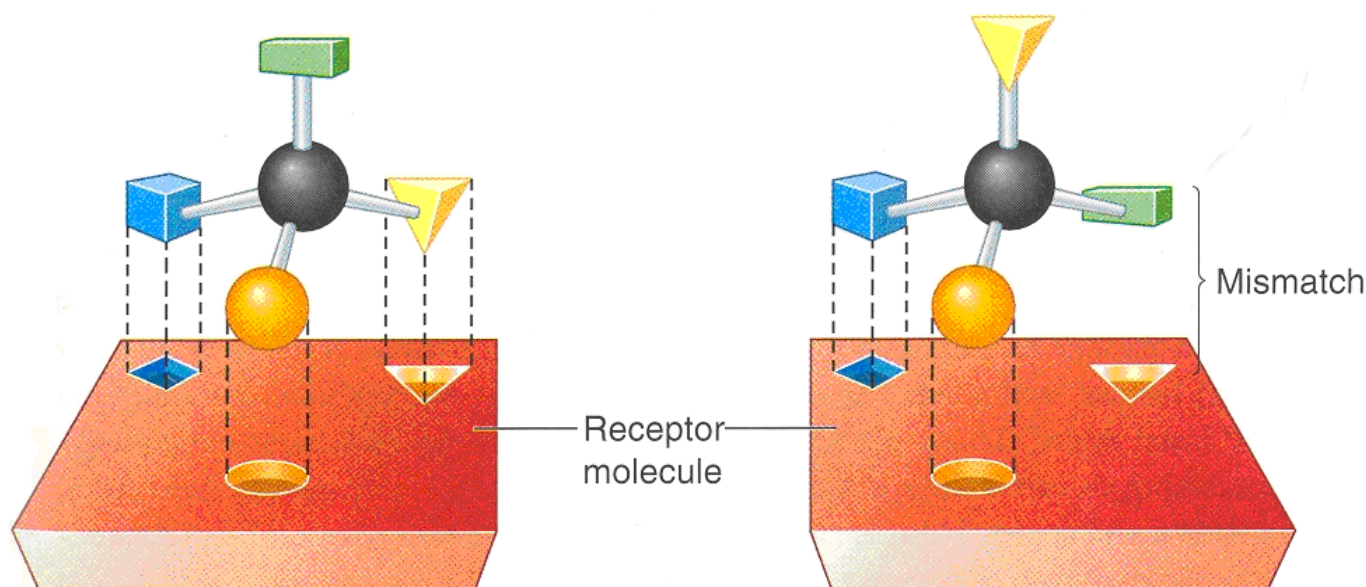


**(S) – 2-Bromobutane**

# Chirality

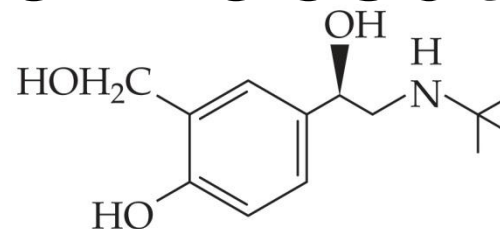
Molecular recognition is carried out by some special biomolecules (e.g. receptors and enzymes) that are themselves spatially arranged in a specific direction.

Most biomolecules possess handedness and their binding sites can thus distinguish between the right-handed or left-handed compounds.

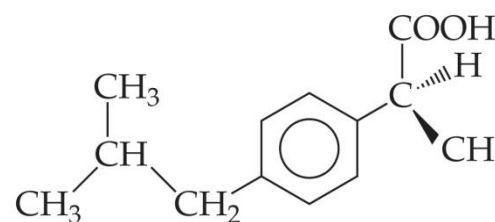
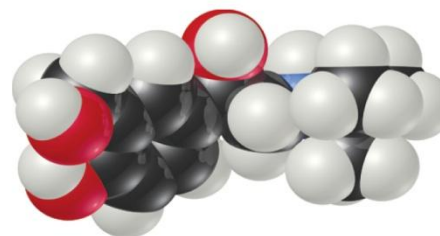


# Chirality and Pharmaceuticals

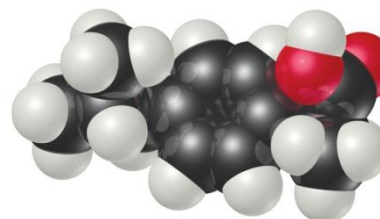
- Many drugs are chiral compounds.
- Equal mixtures of enantiomers is called a racemic mixture. Often only one enantiomer is clinically active; the other can be inert OR harmful.



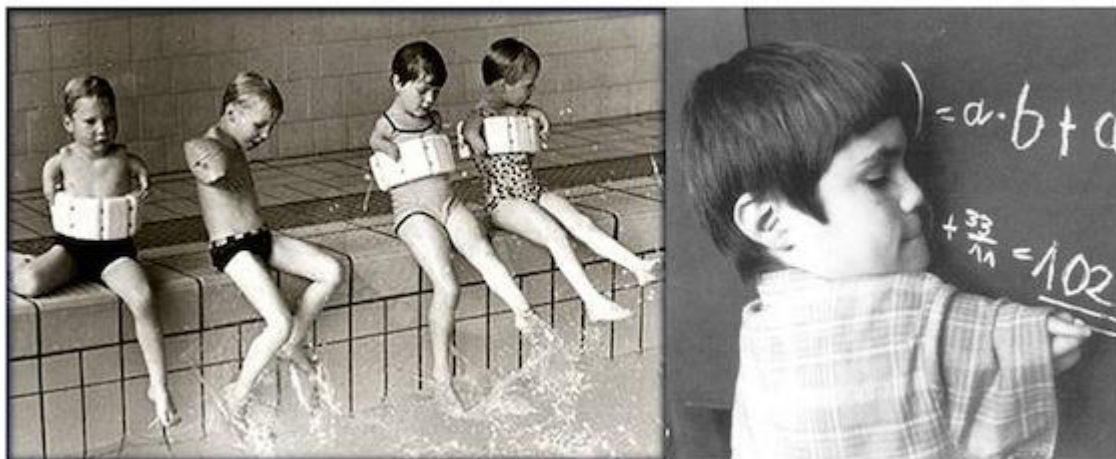
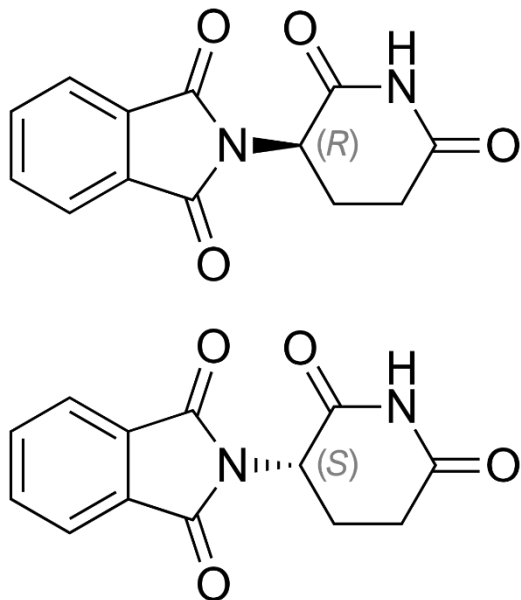
(R)-Albuterol



(S)-Ibuprofen



# Thalidomide



In the 1950's and early 1960's, mothers who had been taking the drug thalidomide for morning sickness began delivering babies with malformed limbs or missing arms and legs.

The drug was proven to cause these defects and pulled in 1961.

- <https://www.acs.org/molecule-of-the-week/archive/t/thalidomide.html>
- <https://cen.acs.org/articles/96/i8/targeted-protein-degraders-are-redefining-how-small-molecules-look-and-act.html>